



Trivyn

Where Formal Reasoning Meets Generative
Intelligence

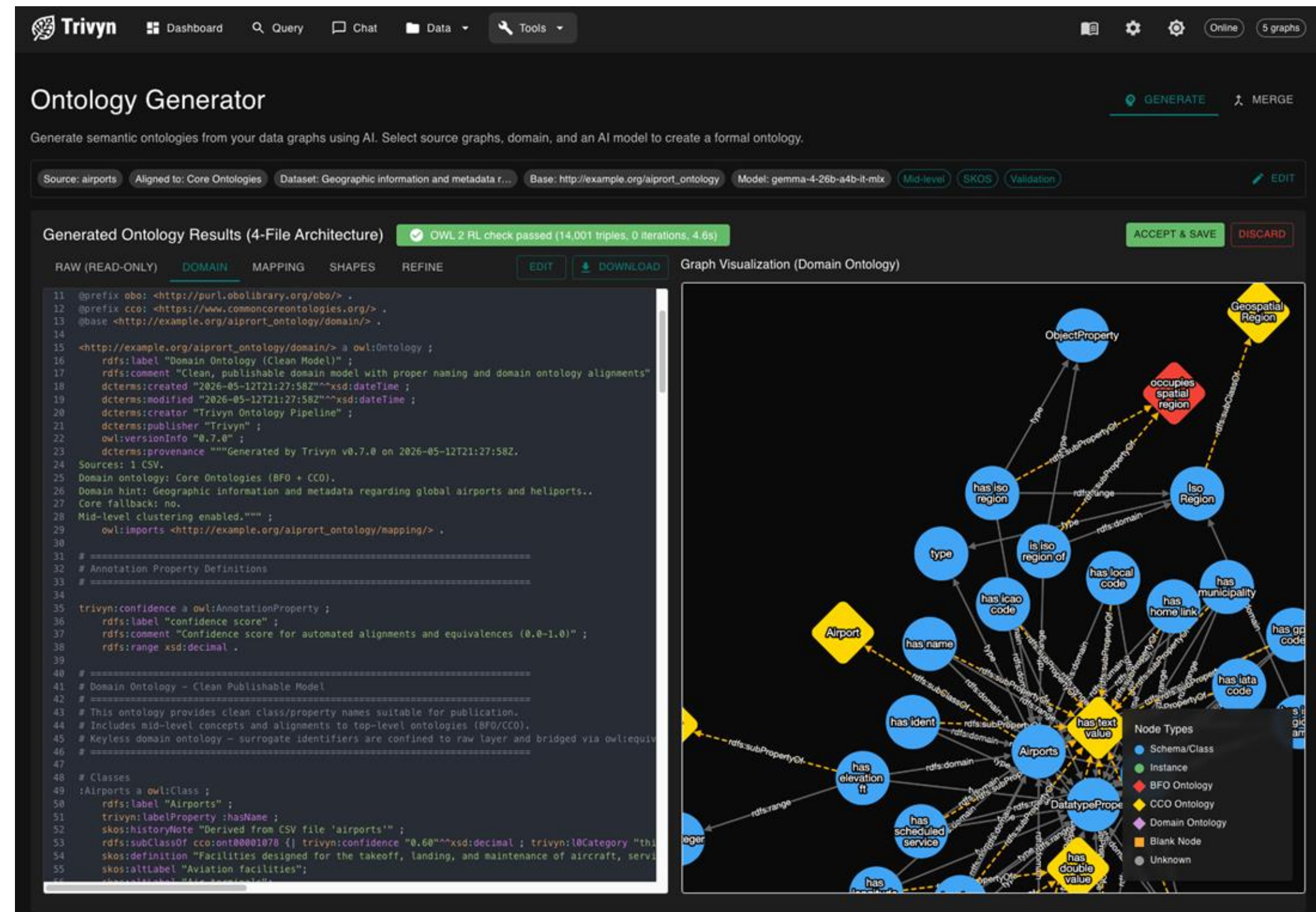
STIDS 2026 | James Adam

Why Trivyn?

I wanted to build:

- A powerful knowledge graph platform that scaled down to a single workstation and was truly easy to deploy
- A modern, intelligent ontology generator that gave me a solid BFO/CCO-aligned starting point
- Trustworthy natural language query capability I could rely on

And I wanted everything to work fully offline with zero cloud dependency or vendor lock-in

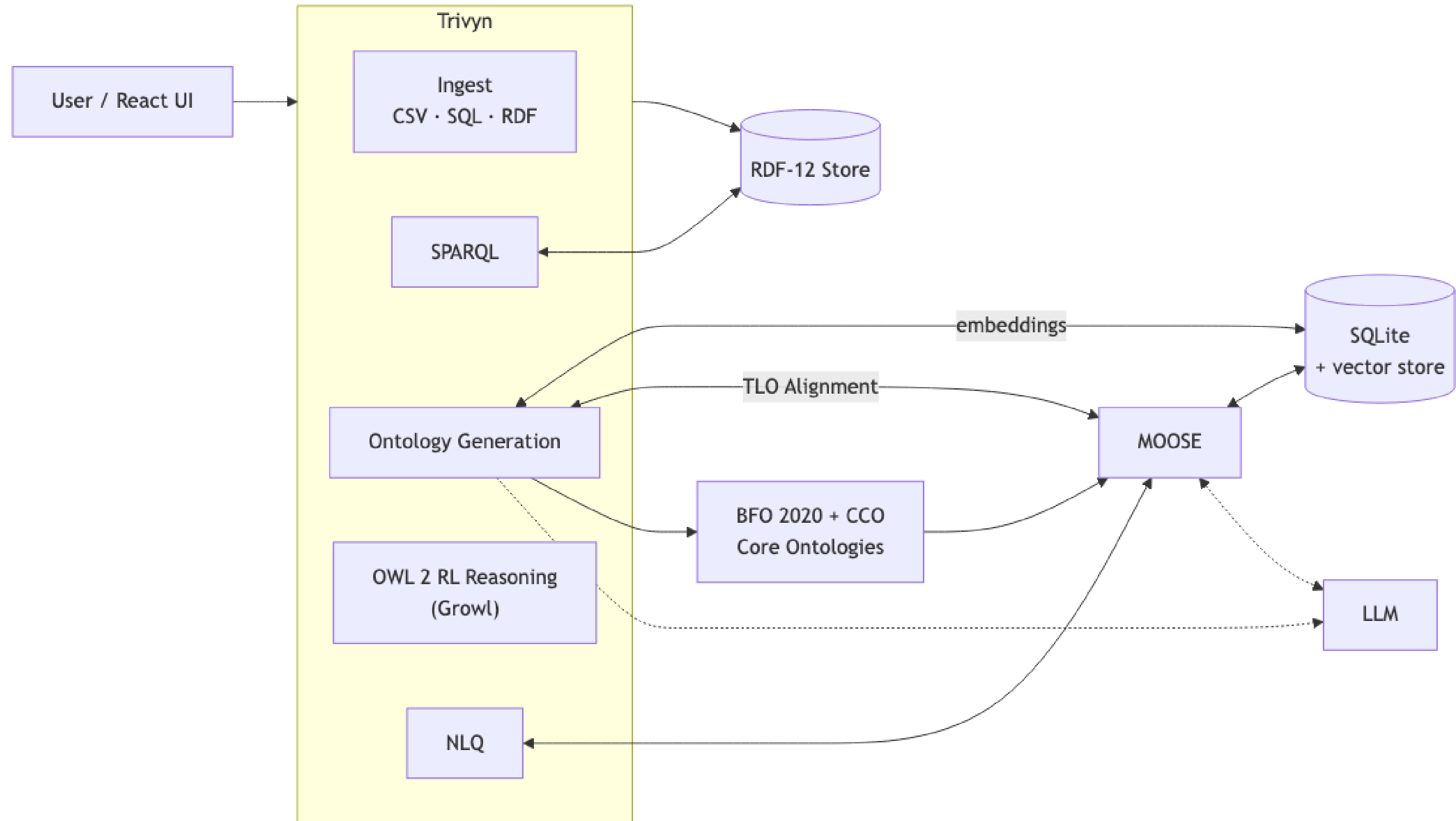


The screenshot displays the Trivyn Ontology Generator interface. The top navigation bar includes links for Dashboard, Query, Chat, Data, and Tools. The main heading is "Ontology Generator" with a subtext: "Generate semantic ontologies from your data graphs using AI. Select source graphs, domain, and an AI model to create a formal ontology." Below this, a configuration bar shows settings: Source: airports, Aligned to: Core Ontologies, Dataset: Geographic information and metadata r..., Base: http://example.org/airport_ontology, Model: gemma-4-26b-a4b-it-mix, Mid-level, SKOS, Validation, and an EDIT button.

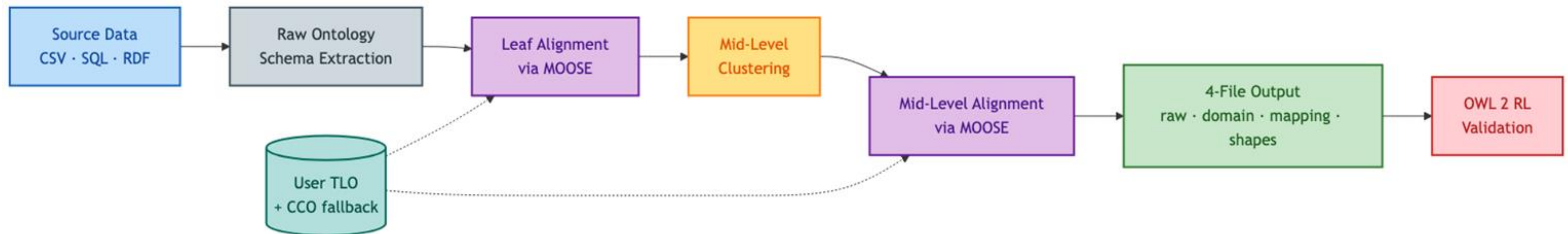
The "Generated Ontology Results (4-File Architecture)" section shows a status bar: "OWL 2 RL check passed (14,001 triples, 0 iterations, 4.6s)" and buttons for ACCEPT & SAVE and DISCARD. Below this, there are tabs for RAW (READ-ONLY), DOMAIN, MAPPING, SHAPES, and REFINE. The DOMAIN tab is active, displaying the generated ontology in Turtle syntax. The graph visualization on the right shows a network of nodes and edges, with a legend for Node Types: Schema/Class (blue circle), Instance (green circle), BFO Ontology (red diamond), CCO Ontology (yellow diamond), Domain Ontology (orange diamond), Blank Node (grey circle), and Unknown (grey circle).

```
11 @prefix obo: <http://purl.obolibrary.org/obo/> .
12 @prefix cco: <https://www.commoncoreontologies.org/> .
13 @base <http://example.org/airport_ontology/domain/> .
14
15 <http://example.org/airport_ontology/domain/> a owl:Ontology ;
16   rdfs:label "Domain Ontology (Clean Model)" ;
17   rdfs:comment "Clean, publishable domain model with proper naming and domain ontology alignments" ;
18   dcterms:created "2026-05-12T21:27:58Z"^^xsd:dateTime ;
19   dcterms:modified "2026-05-12T21:27:58Z"^^xsd:dateTime ;
20   dcterms:creator "Trivyn Ontology Pipeline" ;
21   dcterms:publisher "Trivyn" ;
22   owl:versionInfo "0.7.0" ;
23   dcterms:provenance "Generated by Trivyn v0.7.0 on 2026-05-12T21:27:58Z." ;
24
25 Sources: 1 CSV.
26 Domain ontology: Core Ontologies (BFO + CCO).
27 Domain hint: Geographic information and metadata regarding global airports and heliports..
28 Core fallback: no.
29 Mid-level clustering enabled."" ;
30 owl:imports <http://example.org/airport_ontology/mapping/> .
31
32 # =====
33 # Annotation Property Definitions
34 # =====
35
36 trivyn:confidence a owl:AnnotationProperty ;
37   rdfs:label "confidence score" ;
38   rdfs:comment "Confidence score for automated alignments and equivalences (0.0-1.0)" ;
39   rdfs:range xsd:decimal .
40
41 # =====
42 # Domain Ontology - Clean Publishable Model
43 # =====
44 # This ontology provides clean class/property names suitable for publication.
45 # Includes mid-level concepts and alignments to top-level ontologies (BFO/CCO).
46 # Keyless domain ontology - surrogate identifiers are confined to raw layer and bridged via owl:equivalentClass.
47 # =====
48
49 # Classes
50 :Airports a owl:Class ;
51   rdfs:label "Airports" ;
52   trivyn:labelProperty hasName ;
53   skos:historyNote "Derived from CSV file 'airports'" ;
54   rdfs:subClassOf cco:ont00001078 { | trivyn:confidence "0.60"^^xsd:decimal ; trivyn:l0Category "this is a surrogate identifier for the takeoff, landing, and maintenance of aircraft, serving as a placeholder for the actual class name" ; } ;
55   skos:definition "Facilities designed for the takeoff, landing, and maintenance of aircraft, serving as a placeholder for the actual class name" ;
56   skos:altLabel "Aviation facilities" ;
```

Trivyn Architecture



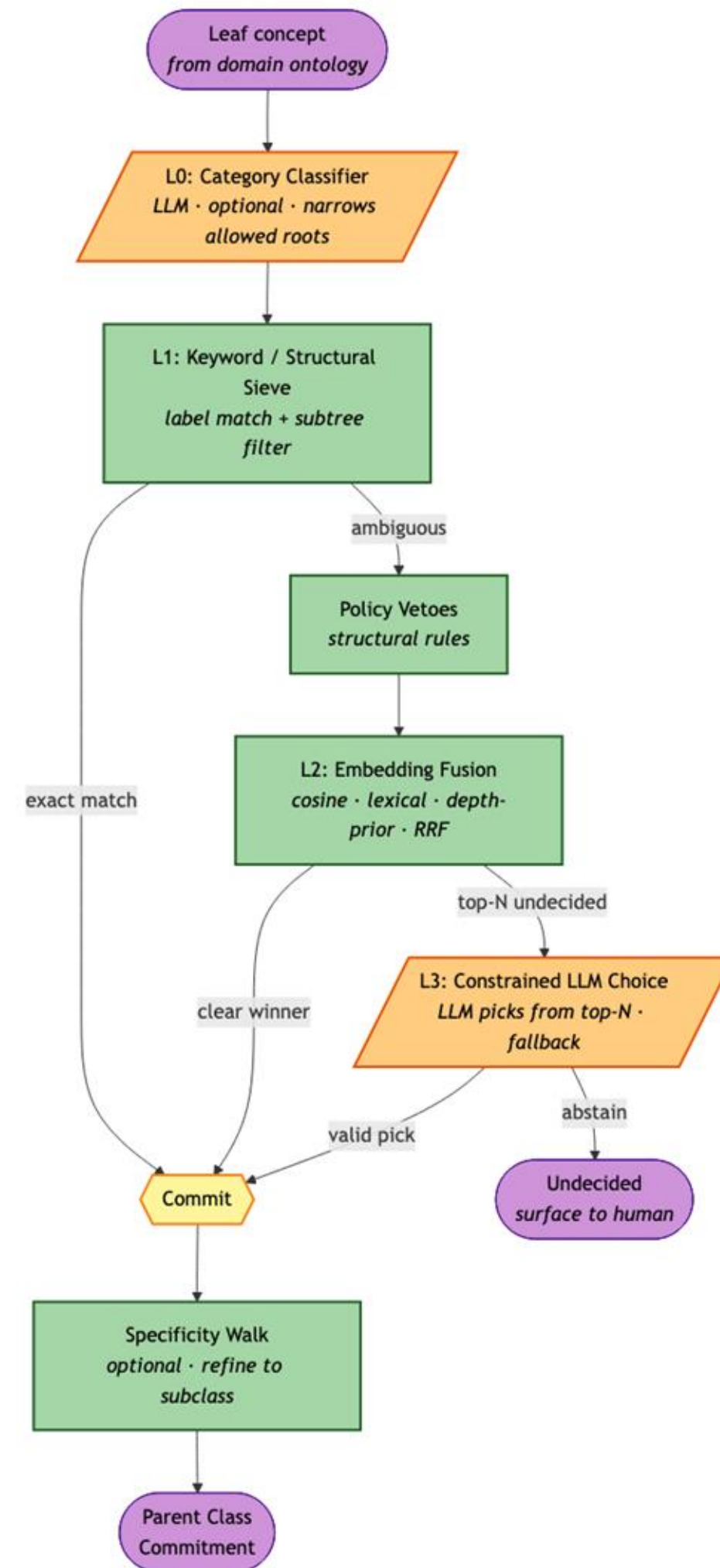
Ontology Generation



- Source specific processors store data patterns in cache on ingest
- The “raw ontology” is generated deterministically using cached data patterns
- Alignment to TLO is handled in MOOSE
- OWL 2 RL validation uses the integrated GROWL reasoner.
 - GROWL - open source reasoner: <https://github.com/trivyn/growl>

TLO alignment (MOOSE)

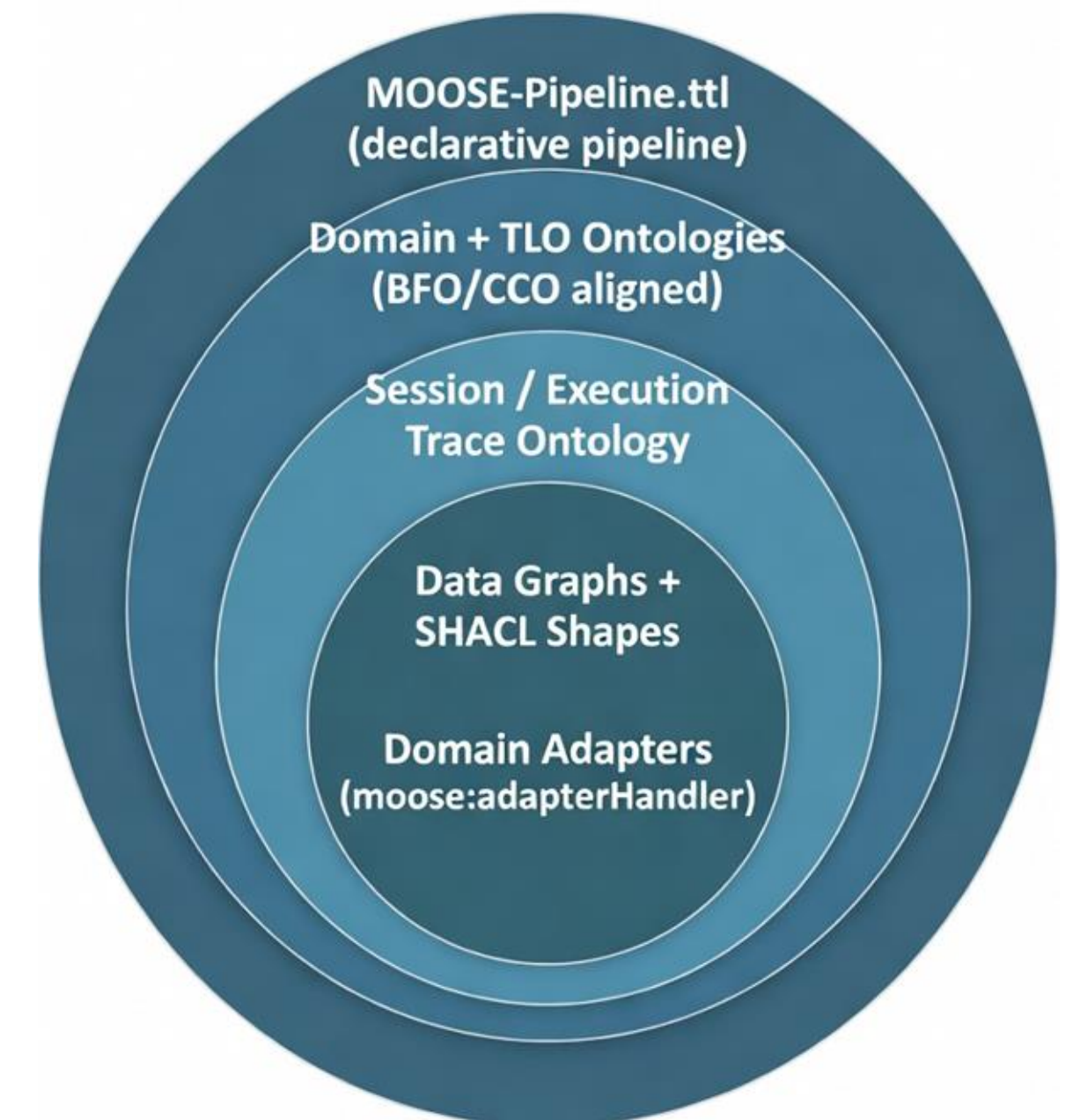
- Four tiers
 - Category classification
 - Keyword / Structural
 - Embedding Fusion
 - Constrained LLM choice
 - LLM picks from a curated list
 - Specificity walk: post-commit, descend subClassOf chain toward more specific subclasses while label-cosine improves



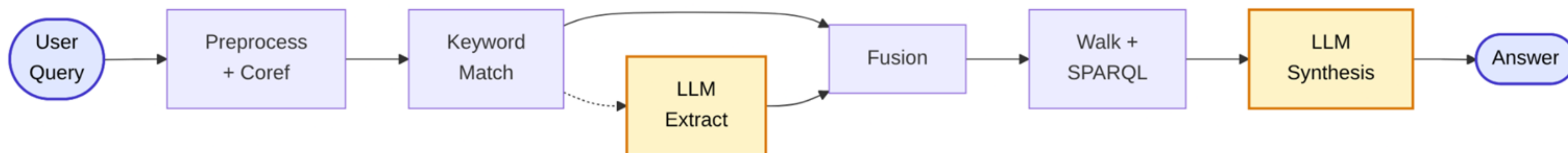
Modular Ontology-Orchestrated Synthesis Engine

- Pipeline, knowledge model, session state, and extensions are all ontological
- Richer ontology annotations can substitute for larger LLMs.
- Symbolic layer does the heavy lifting
- LLM operates only as a narrow NL sensor

Ontologies All the Way Down



MOOSE Query Pipeline



- Hybrid neurosymbolic design
 - Symbolic layer does the heavy lifting, LLM acts only as a narrow sensor
- Keyword match first
 - fast deterministic path; LLM extraction is only triggered when needed
- Single fusion point + ontology driven walk
- LLM synthesis only at the end
 - final answer formatting, never query planning or reasoning

Demo

Trivyn

Dashboard

Query

Chat

Data

Tools

Online

13 graphs

Data Management

Import data from files and databases, export graphs, and monitor history

IMPORTEXPORTHISTORY

Format

JSON

Import JSON Data

Upload JSON files and convert them to RDF. Nested objects become linked entities.

airports.json

0.00 MB

Use Existing Graph (optional)

Select an existing graph to add data to

OR CREATE NEW GRAPH

Target Graph Configuration

Target Graph IRI *

http://example.org/json/airports

The named graph where the RDF data will be stored

Short Name (optional)

Airports JSON Data

A friendly display name for this graph

Description (optional)

Airport Data

A detailed description of what this graph contains

IMPORT JSON DATA

Screenshot

Trivyn

Dashboard

Query

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Online14 graphs

Ontology Generator

GENERATEMERGE

Generate semantic ontologies from your data graphs using AI. Select source graphs, domain, and an AI model to create a formal ontology.

Source Graphs

Select Graphs

Airport JSON Data

1 graph(s) selected • Core Ontologies • Mid-level enabled (max: 50)

Domain Ontology

Domain Ontology

Core Ontologies (BFO + CCO)1699 embeddings

Dataset description (optional)

This dataset contains information about various international airports, including

AUTO-DETECT

One sentence describing the dataset's domain.
Improves definitions. 108/300

Base IRI

http://example.org/airports_ontology

Namespace for generated ontology group

Generation Options

Mid-Level Ontology

Max Concepts50Min Cluster2

Generates intermediate classes using similarity

SKOS Annotation Enrichment

AI generates definitions, synonyms, examples

Fast mode (skip property annotations)

Full enrichment including per-property annotations

LLM Semantic Validation

AI validates ambiguous domain assignments

AI Model

Select Model

gemma-4-e4b-it-mixby organization_owner

GENERATE

Ready to Generate Ontology

Select source graphs, application domain, and an AI model, then click "Generate Ontology" to create a semantic ontology from your data.

Screenshot

Ontology Generator

Generate semantic ontologies from your data graphs using AI. Select source graphs, domain, and an AI model to create a formal ontology.

GENERATE

MERGE

Source: airports_complete

Aligned to: Core Ontologies

Dataset: This dataset contains comprehensive g...

Base: http://example.org/airports_ontology_2

Model: gemma-4-e4b-it-mlx

Mid-level

SKOS

Validation

EDIT

Generated Ontology Results (4-File Architecture)

OWL 2 RL check passed (14,038 triples, 0 iterations, 4.4s)

ACCEPT & SAVE

DISCARD

RAW (READ-ONLY)

DOMAIN

MAPPING

SHAPES

REFINE

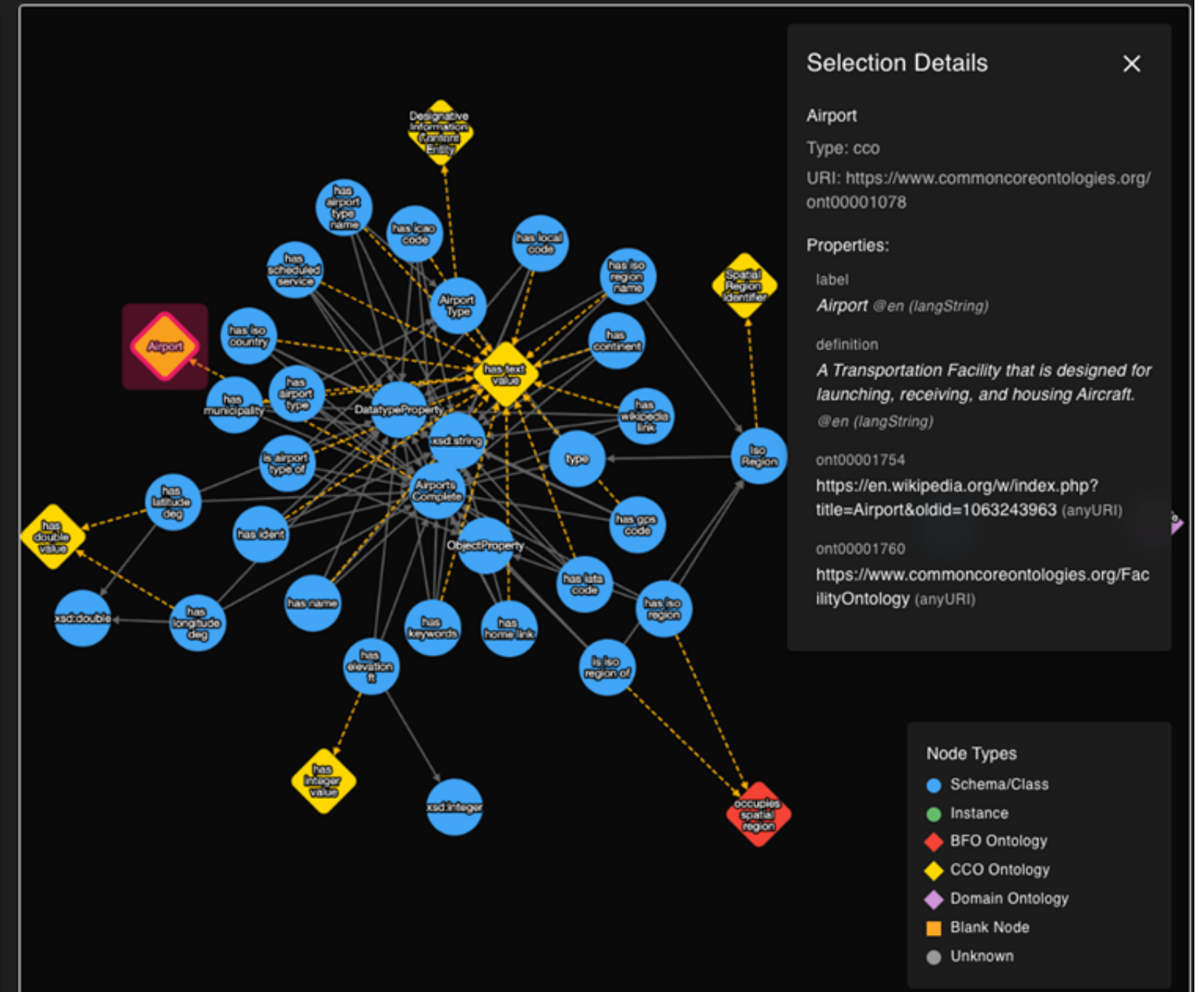
BINDINGS

EDIT

DOWNLOAD

```
29 owl:imports <http://example.org/airports_ontology_2/mapping/> .
30
31 # =====
32 # Annotation Property Definitions
33 # =====
34
35 trivyn:confidence a owl:AnnotationProperty ;
36   rdfs:label "confidence score" ;
37   rdfs:comment "Confidence score for automated alignments and equivalences (0.0~1.0)" ;
38   rdfs:range xsd:decimal .
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45 # Keyless domain ontology - surrogate identifiers are confined to raw layer and bridged via owl:equivalentClass/Property axi
46 # =====
47
48 # Classes
49 :AirportsComplete a owl:Class ;
50   rdfs:label "Airports Complete" ;
51   trivyn:labelProperty :hasName ;
52   skos:historyNote "Derived from CSV file 'airports_complete'" ;
53   rdfs:subClassOf cco:ont00001078 { | trivyn:confidence "0.75"^^xsd:decimal ; trivyn:l0Category "thing" ; trivyn:parentLabel
54   skos:definition "A comprehensive record describing a specific airport facility, including its geographical location, ope
55   skos:altLabel "Airport Profile";
56   skos:hiddenLabel "AirportRecord";
57   skos:example "John F. Kennedy International Airport".
58
59 # Lifted Classes (promoted from literal columns to entities)
60 :AirportType a owl:Class ;
61   rdfs:label "Airport Type" ;
62   skos:historyNote "Lifted from column 'AirportType' in table 'AirportsComplete' (low cardinality)" ;
63   trivyn:labelProperty :hasAirportTypeName ;
64   rdfs:subClassOf cco:ont0000686 { | trivyn:confidence "0.65"^^xsd:decimal ; trivyn:l0Category "designator" ; trivyn:paren
65   skos:definition "A classification describing the operational nature or physical characteristics of an airport facility."
66   skos:altLabel "Airport Classification";
67   skos:hiddenLabel "AType";
68   skos:example "balloonport";
69   skos:example "closed";
70   skos:example "heliport";
71   skos:example "large_airport";
72   skos:example "medium_airport";
73   skos:example "seaplane_base";
74   skos:example "small_airport".
75
76 :hasAirportTypeName a owl:DatatypeProperty ;
77   rdfs:label "has airport type name" ;
```

Graph Visualization (Domain Ontology)



Ontology Generator

Generate semantic ontologies from your data graphs using AI. Select source graphs, domain, and an AI model to create a formal ontology.

Source: airports_complete Aligned to: Core Ontologies Dataset: This dataset contains global airport ... Base: http://example.org/airport_complete_ontology Model: gemma-4-e2b-it-mix Mid-level SKOS Validation EDIT

Generated Ontology Results (4-File Architecture) OWL 2 RL check passed (14,063 triples, 0 iterations, 4.4s) ACCEPT & SAVE DISCARD

RAW (READ-ONLY) DOMAIN MAPPING SHAPES **REFINE** BINDINGS

▶ APPLY CHANGES TO TTL FILES ↺ RESET ↻ RE-PARSE 🔗

CLASSES (3)			PROPERTIES (22)		+ NEW CLASS
Domain Name	Raw Name	Label Prop	Props	Status	
Airports	airports_complete	hasName	18	good	
AirportType	AirportType	hasAirportTypeName	2	good	
IsoRegion	IsoRegion	hasIsoRegionName	2	good	

AirportType good ×

BASIC INFO

LABEL PROPERTY

ANNOTATIONS

PROPERTIES

HIERARCHY

BINDING

Annotation type

skos:altLabel — Alternative label

Value

+

Alt Labels (skos:altLabel)

Airport Category

Hidden Labels (skos:hiddenLabel)

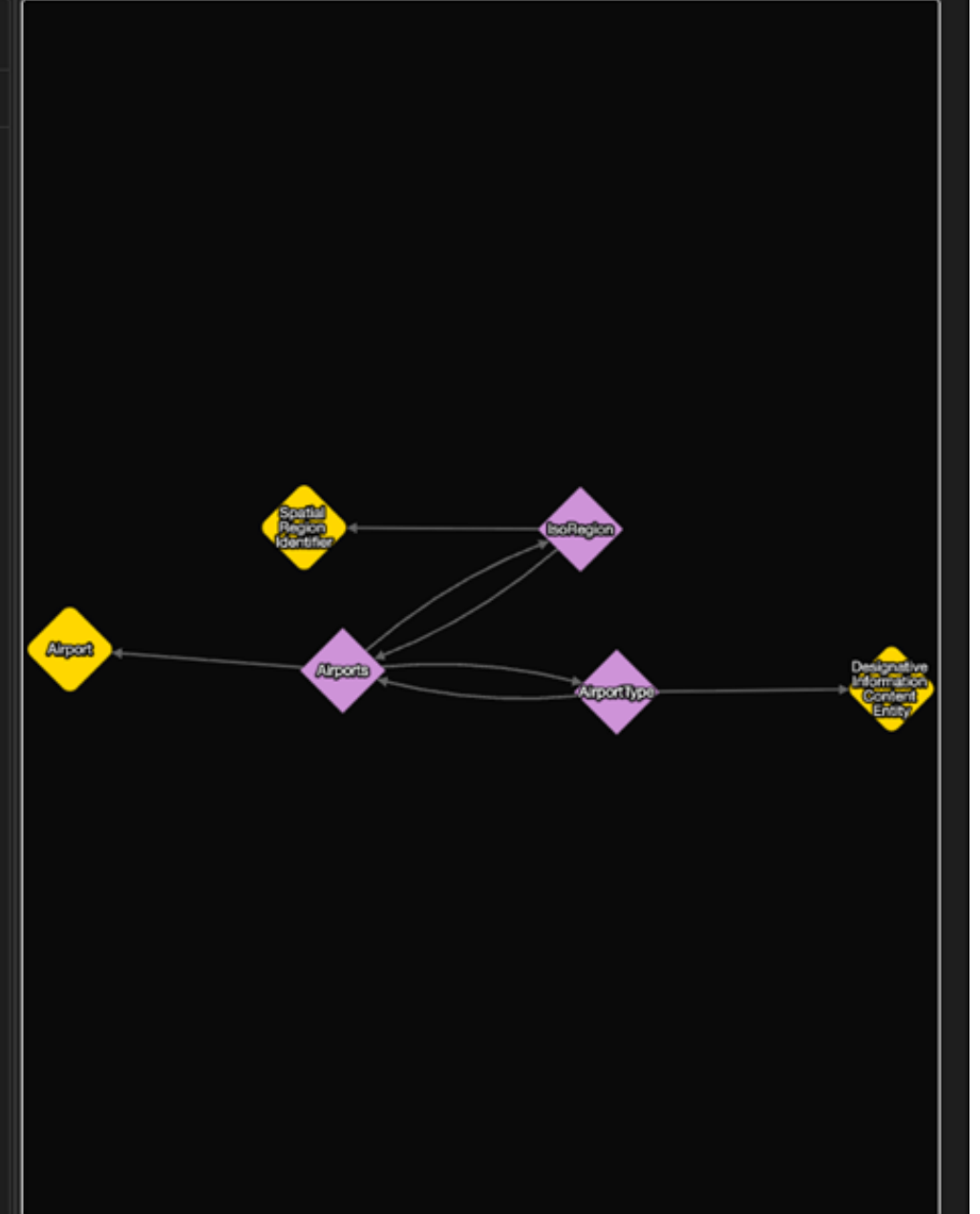
AT

Comments (rdfs:comment)

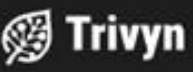
None

Scope Note (skos:scopeNote)

None



Screenshot



Dashboard
Query
Chat
Data
Tools

Online
28 graphs

SESSIONS

Tell me about Etame...
20h ago · 3 turns

Show them to me
23h ago · 2 turns

AI Model

gemma-4-26b-a4b-it-mix
by organization_owner

NEW

Graphs

WPI Aligned Data

Query will be executed against 1 selected graph(s)

LLM Assist Level

Symbolic

Standard

Relaxed

Planning

Validation

Fallback

Ask for clarification on unresolved terms

Training mode (suppress auto-resolution)

Focus

Etame Terminal
Tema
Ain Sukhna Terminal
Serpentina Terminal
Al Shaheen Terminal

Communication capabilities are extensive, including airport communications, radio, radiotelephone, rail, and telephone. The port also provides medical facilities and quarantine sanitation services. Navigation information can be found in the Sailing Directions Pub. 123 (Enroute) - Southwest Coast of Africa, and digital nautical charts are available under identifiers a0146065, coa01a, gen01b, and h0146070. Note that there are entrance restrictions regarding tides and heavy swells, and the port has a US Representative.

Caveat: Extraction confidence was 45%.

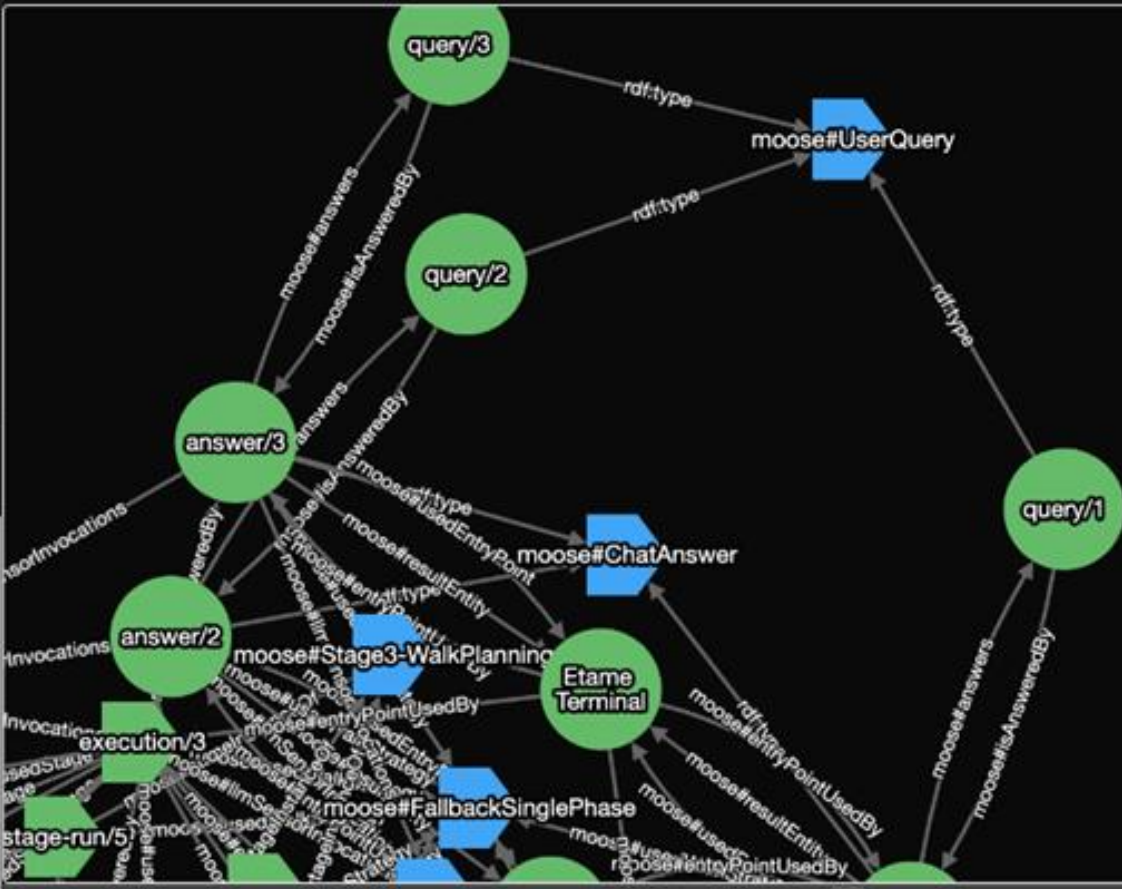
Tell me about Etame Terminal

GRAPH

MAP

DATA

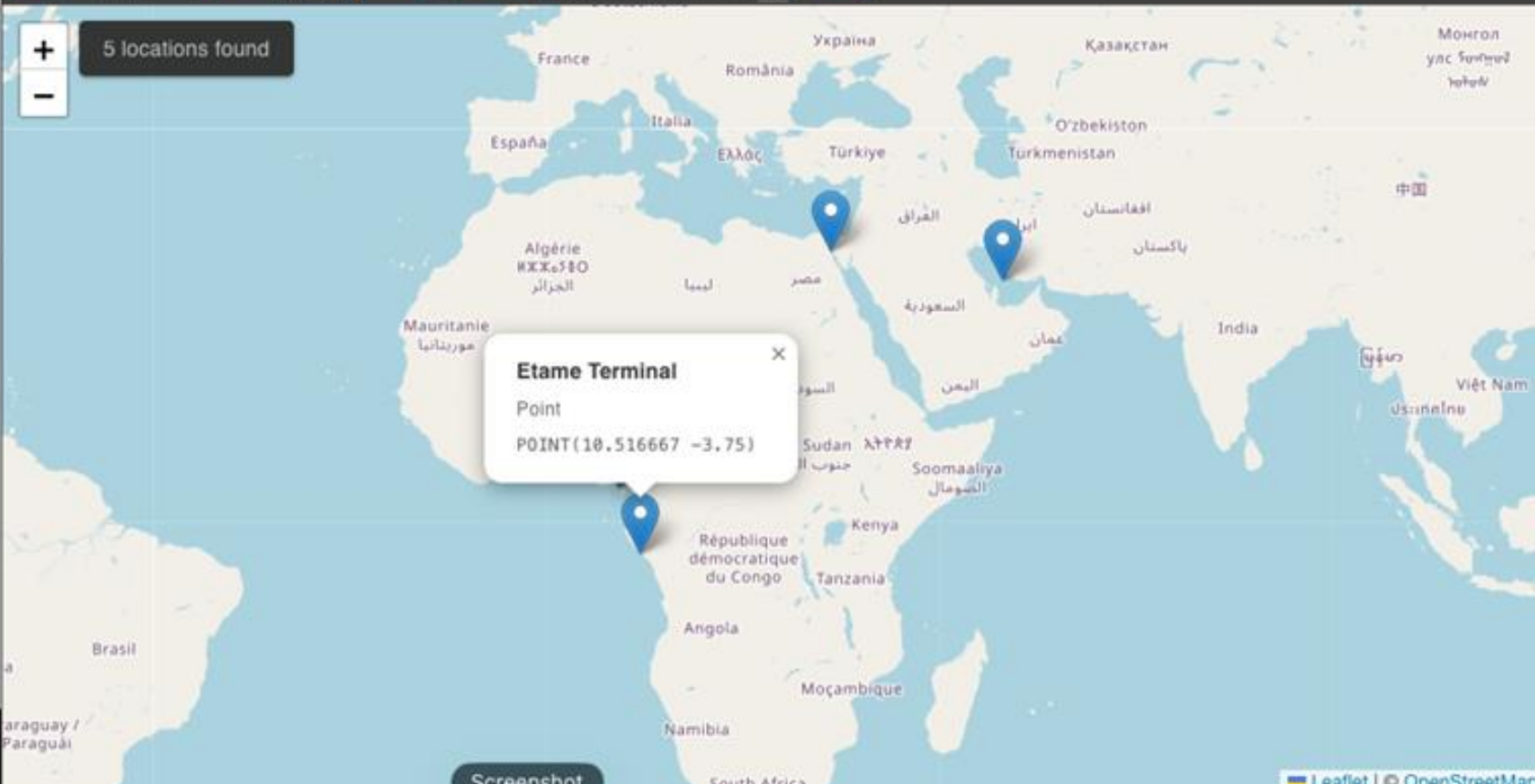
GRAPH + MAP



Node Types

- Schema/Class
- Instance
- BFO Ontology
- CCO Ontology
- Domain Ontology
- Blank Node
- MOOSE Ontology
- Execution Trace
- Unknown

+
-
5 locations found



Etame Terminal
Point
POINT(10.516667 -3.75)

Etame Terminal, also known as FPSO Petroleo Nautipa, is a very small port located in Gabon (Region: Gabon -- 46410) within the South Atlantic Ocean. It is classified as an Open Roadstead harbor type and provides poor shelter. The terminal's UNLOCODE is GA ETA.

The terminal features significant maritime depths, including an anchorage depth of 76.0 meters, a cargo pier depth of 23.2 meters, and a channel depth of 23.2 meters. The oil terminal depth is also recorded at 76.0 meters. While the entrance width, maximum vessel length, beam, draft, and offshore maximum vessel dimensions are listed as 0.0 or not specified, the terminal does include a turning area.

Navigation and communication services are available at the terminal. Pilotage is compulsory and available, and there is an estimated time of arrival message service. Communication capabilities include radio, telephone, and telefax, though there is no rail communication. The terminal does not have entrance restrictions related to tides or ice, but it does have entrance restrictions regarding heavy swells.

Technical and navigational data for the terminal includes digital nautical charts (coa01h, gen01b) and a standard nautical chart (57220.0). Further information can be found in the Sailing Directions Pub. 123 (Enroute) - Southwest Coast of Africa. The terminal's location is identified by the coordinates POINT(10.516667 -3.75).

Note: Extraction confidence is 70% — All entities and relationships resolved cleanly.

Ask about your knowledge graph...

AI Model

gemma-4-e4b-it-mlx

by organization_owner

NEW

Graphs

Aligned Airport Data

Query will be executed against 1 selected graph(s)

LLM Assist Level

Symbolic

Standard

Relaxed

Planning

Validation

Fallback

Ask for clarification on unresolved terms

Training mode (suppress auto-resolution)

How many airfields are there?

I couldn't identify any specific entities in your query. What did you mean — can you point at a class or instance in the data?

Pick an entity:

Airports

Remember for me

DECLINE

Pick: <http://example.org/airport_ontology/domain/Airports>

There are 30 airfields.

Trivyn

Dashboard

Query

Chat

Data

Tools

Online

28 graphs

Pending Approvals

User-supplied clarifications waiting for ontologist review. Approve to promote into the canonical ontology; reject to revoke.

User staging: 1

Session-only: 0

Rejected — removed 1 triple(s)

- Airport Ontology

http://example.org/airport_ontology/domain/

Scope	Target	Predicate	Surface text	Confidence	Created	Actions
user	Airports	altLabel	"airfields"	0.75	5/25/2026, 8:10:49 AM	✓ ✗

- Training mode disables LLM resolution assistance to find gaps in the ontology
- User clarifications are sent to an approval queue for review before being promoted to the ontology